Lab Report: MSFT RNN Stock Prediction

Course: SP26 BFOR516 - Adv Data Analytics for Cyber

Instructor: Srishti Gupta, Ph.D.

Dataset: Microsoft (MSFT) Historical Stock Data (NASDAQ)

Student: Spencer Kone

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1. AI Disclosure

During this lab, Antigravity AI (Google DeepMind) served as a programming collaborator. The AI provided support across several project components:

1. Creating the basic Jupyter Notebook framework and developing standard code with pandas, matplotlib, and sklearn.preprocessing.MinMaxScaler.
2. Resolving a Python 3.13 compatibility problem by upgrading the environment to TensorFlow 2.21, which allowed the notebook to run correctly.
3. Developing matplotlib visualization code to compare SimpleRNN and LSTM model performance effectively.
4. Auditing the model architecture, training, and code..

I carefully reviewed, processed, verified, and cognitively understood all AI-produced code and analysis before incorporating it into the project. The analysis, explanations, and modeling approach reflect my genuine comprehension of Recurrent Neural Network.

2. Lab Objective

This lab concentrated on building a Recurrent Neural Network (RNN) through the TensorFlow/Keras framework to forecast Microsoft (MSFT) closing stock prices. The methodology mirrored the classroom demonstration using AAPL data, integrating data normalization, sliding window sequence creation, and sequential prediction modeling techniques adapted for the MSFT dataset.

3. Workflow & Preprocessing

- Data Import: The MSFT dataset, spanning 125 trading days, was loaded into a Pandas DataFrame.

- Data Cleaning: Price data underwent cleaning by removing dollar signs ($) and converting values to floating-point format. Date information was processed into datetime objects and sorted chronologically from oldest to newest.

- Data Normalization: Given that RNNs rely on gradient descent and utilize tanh/sigmoid activation functions, closing prices were scaled to a [0, 1] range using MinMaxScaler.

- Sequence Generation: A sliding window approach was implemented with a look_back period of 10 days, equivalent to two standard trading weeks. This duration strikes a balance between capturing short-term market behaviors and maintaining sensitivity to recent trends. The model was trained to use 10 consecutive days of historical data to forecast the closing price for day 11. Chronological integrity was preserved by allocating the first 80% of data for training and the final 20% for testing, preventing future data leakage into the training process.

4. Model Architectures & Training

- Two distinct models were constructed for performance comparison:
- SimpleRNN Model: This sequential architecture featured a SimpleRNN layer with 64 units configured to return sequences, a Dropout layer set to 20% for overfitting prevention, an additional SimpleRNN layer with 32 units, and a final Dense layer for output generation.

- LSTM Model: This design addresses the vanishing gradient problem inherent in SimpleRNNs. While maintaining the same structural framework, it substituted standard RNN layers with Long Short-Term Memory (LSTM) cells that employ gated memory mechanisms.
- Both models were configured using the Adam optimizer and Mean Squared Error (MSE) loss function. Overfitting prevention was achieved through Keras EarlyStopping, which monitored validation loss with 15 epochs of patience, utilizing a chronological 15% validation split from the training data's end. This

strategy preserved the integrity of the 20% test set by maintaining complete separation during the training phase.

5. Evaluation & Results

Following prediction generation, normalized [0, 1] model outputs were converted back to actual USD values through the scaler's inverse_transform() method. Both models demonstrated robust performance on test data, successfully following the general volatile trends in MSFT stock prices from January through March 2026.

Test Set Performance:

Note: A random seed (seed=42) was established at the notebook's beginning to ensure consistent and reproducible results across all reported outputs and models.

• SimpleRNN: Successfully captured short-term directional movements but occasionally displayed heightened sensitivity to abrupt daily price fluctuations.

• LSTM: Demonstrated superior memory retention across extended sequences. Since SimpleRNNs employ Backpropagation Through Time (BPTT), they face challenges with longer sequences due to the vanishing gradient problem outlined in lecture slides 23-27. The LSTM's internal gating mechanisms effectively resolved this mathematical constraint, yielding smoother predictions and improved error metrics including reduced RMSE and MAPE values.

6. Future Price Forecast

The best-performing model was employed to predict the subsequent 5 trading days beyond the dataset's final date.

Using a recursive methodology, the model generated a prediction for day T+1, incorporated this prediction into the sliding window, then utilized this updated window to forecast day T+2, continuing this sequence through T+5. The outcomes, displayed in the notebook's final output and the msft_forecast.png visualization, present a realistic extension of existing stock price patterns with appropriate smoothing.

Recursive Prediction Constraints: A significant challenge with this autoregressive method, as referenced in lecture slide 7, involves error propagation. Since each predicted day serves as input for subsequent predictions, minor errors compound over time, substantially diminishing confidence in extended forecasts.

7. Analysis & Conclusion

• Data Limitations: The primary constraint was the relatively modest dataset of 125 days, while deep learning models generally achieve optimal performance with thousands of data points.

• Univariate Approach: The model was designed as a single-variable system that predicted closing prices based exclusively on historical closing price sequences.

• Principal Findings: Despite these limitations, the lab illustrated that fundamental RNN architectures can effectively recognize robust patterns within financial time series data. Consistent with the pattern observed in our classroom AAPL demonstration, the LSTM network successfully overcame the memory constraints of SimpleRNNs, establishing itself as a capable architecture for sequential regression applications.

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8. Code Output

TensorFlow version : 2.21.0
NumPy version : 2.4.1
Pandas version : 2.3.3

Shape: (124, 6)

	Date	Close/Last	Volume	Open	High	Low
0	03/11/2026	$404.88	25512140	$405.57	$409.01	$401.59
1	03/10/2026	$405.76	31706380	$410.03	$410.20	$402.93
2	03/09/2026	$409.41	30131850	$404.915	$410.21	$403.50
3	03/06/2026	$408.96	31123860	$409.20	$413.05	$408.5101
4	03/05/2026	$410.68	39001320	$404.42	$411.61	$404.40
5	03/04/2026	$405.20	35807990	$401.265	$411.0326	$400.31
6	03/03/2026	$403.93	38199210	$393.14	$406.70	$392.67
7	03/02/2026	$398.55	35474910	$392.855	$401.19	$390.63
8	02/27/2026	$392.74	51367200	$390.88	$396.82	$389.88
9	02/26/2026	$401.72	34405870	$404.71	$407.49	$398.74

Columns: ['Date', 'Close/Last', 'Volume', 'Open', 'High', 'Low']

Data types:
Date object
Close/Last object
Volume int64
Open object
High object
Low object
dtype: object

Missing values:
Date 0
Close/Last 0
Volume 0
Open 0
High 0
Low 0

dtype: int64

Descriptive Statistics (raw):

	Date	Close/Last	Volume	Open	High	Low
count	124	124	1.240000e+02	124	124	124
unique	124	123	NaN	124	124	122
top	03/11/2026	$483.47	NaN	$405.57	$409.01	$475.86
freq	1	2	NaN	1	1	2
mean	NaN	NaN	2.795272e+07	NaN	NaN	NaN
std	NaN	NaN	1.491522e+07	NaN	NaN	NaN
min	NaN	NaN	5.855880e+06	NaN	NaN	NaN
25%	NaN	NaN	1.886896e+07	NaN	NaN	NaN
50%	NaN	NaN	2.459818e+07	NaN	NaN	NaN
75%	NaN	NaN	3.403432e+07	NaN	NaN	NaN
max	NaN	NaN	1.288553e+08	NaN	NaN	NaN

Cleaned DataFrame — first 5 rows:

	Date	Close	Volume	Open	High	Low
0	2025-09-12	509.90	23624880	506.65	512.55	503.85
1	2025-09-15	515.36	17143790	508.79	515.47	507.00
2	2025-09-16	509.04	19711920	516.88	517.23	508.60
3	2025-09-17	510.02	15816590	510.62	511.29	505.93
4	2025-09-18	508.45	18913700	511.49	513.07	507.66

Date range: 2025-09-12 → 2026-03-11
Total trading days: 124

<Figure size 1400x800 with 2 Axes>

Figure saved as msft_eda.png

Original price range: $384.47 – $542.07
Scaled price range : 0.0000 – 1.0000

Sequence dataset: X.shape=(114, 10, 1), y.shape=(114,)

Training samples : 91

Testing samples : 23

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

[1mModel: "SimpleRNN_Model"[0m

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┏━━━
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[1m Total params: [0m[38;5;34m7,361[0m (28.75 KB)

[1m Trainable params: [0m[38;5;34m7,361[0m (28.75 KB)

[1m Non-trainable params: [0m[38;5;34m0[0m (0.00 B)

Epoch 1/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m11s[0m 1s/step - loss: 0.2538 - mae: 0.4391

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m2s[0m 31ms/step - loss: 0.2073 - mae: 0.3691 - val_loss: 0.0950 - val_mae: 0.2831

Epoch 2/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.1236 - mae: 0.2761

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.1800 - mae: 0.3301 - val_loss: 0.1203 - val_mae: 0.3212

Epoch 3/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0746 - mae: 0.2346

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.1630 - mae: 0.3271 - val_loss: 0.0236 - val_mae: 0.1075

Epoch 4/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 25ms/step - loss: 0.3242 - mae: 0.4896

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.1560 - mae: 0.3110 - val_loss: 0.0456 - val_mae: 0.1960

Epoch 5/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.3706 - mae: 0.5183

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.1838 - mae: 0.3501 - val_loss: 0.0256 - val_mae: 0.1285

Epoch 6/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0883 - mae: 0.2555

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.1241 - mae: 0.2809 - val_loss: 0.1272 - val_mae: 0.3366

Epoch 7/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.1692 - mae: 0.3375

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.1315 - mae: 0.3000 - val_loss: 0.0181 - val_mae: 0.1109

Epoch 8/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 28ms/step - loss: 0.0663 - mae: 0.2127

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0565 - mae: 0.1934 - val_loss: 0.0481 - val_mae: 0.1875

Epoch 9/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 27ms/step - loss: 0.0242 - mae: 0.1481

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0583 - mae: 0.1893 - val_loss: 0.0299 - val_mae: 0.1382

Epoch 10/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0508 - mae: 0.1846

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0708 - mae: 0.2123 - val_loss: 0.0156 - val_mae: 0.1025

Epoch 11/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.1191 - mae: 0.2967

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0485 - mae: 0.1767 - val_loss: 0.0259 - val_mae: 0.1313

Epoch 12/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 24ms/step - loss: 0.0456 - mae: 0.1516

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0529 - mae: 0.1808 - val_loss: 0.0304 - val_mae: 0.1397

Epoch 13/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 28ms/step - loss: 0.0380 - mae: 0.1751

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0400 - mae: 0.1580 - val_loss: 0.0167 - val_mae: 0.1092

Epoch 14/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0268 - mae: 0.1379

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0470 - mae: 0.1647 - val_loss: 0.0175 - val_mae: 0.1120

Epoch 15/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0471 - mae: 0.1646

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0351 - mae: 0.1501 - val_loss: 0.0168 - val_mae: 0.1071

Epoch 16/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0616 - mae: 0.2013

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0349 - mae: 0.1426 - val_loss: 0.0213 - val_mae: 0.1186

Epoch 17/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 19ms/step - loss: 0.0117 - mae: 0.0928

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0355 - mae: 0.1440 - val_loss: 0.0215 - val_mae: 0.1169

Epoch 18/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0317 - mae: 0.1473

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0340 - mae: 0.1507 - val_loss: 0.0224 - val_mae: 0.1189

Epoch 19/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0292 - mae: 0.1243

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0229 - mae: 0.1183 - val_loss: 0.0162 - val_mae: 0.1042

Epoch 20/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0297 - mae: 0.1312

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0328 - mae: 0.1344 - val_loss: 0.0240 - val_mae: 0.1220

Epoch 21/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0333 - mae: 0.1493

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0291 - mae: 0.1367 - val_loss: 0.0196 - val_mae: 0.1116

Epoch 22/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0333 - mae: 0.1717

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0346 - mae: 0.1524 - val_loss: 0.0126 - val_mae: 0.0883

Epoch 23/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0976 - mae: 0.2597

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0357 - mae: 0.1434 - val_loss: 0.0264 - val_mae: 0.1271

Epoch 24/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 19ms/step - loss: 0.0253 - mae: 0.1515

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0268 - mae: 0.1345 - val_loss: 0.0126 - val_mae: 0.0877

Epoch 25/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0737 - mae: 0.2274

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0477 - mae: 0.1851 - val_loss: 0.0138 - val_mae: 0.0872

Epoch 26/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 17ms/step - loss: 0.0365 - mae: 0.1688

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0289 - mae: 0.1369 - val_loss: 0.0134 - val_mae: 0.0858

Epoch 27/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0201 - mae: 0.1222

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0308 - mae: 0.1330 - val_loss: 0.0130 - val_mae: 0.0965

Epoch 28/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 17ms/step - loss: 0.0526 - mae: 0.1893

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0247 - mae: 0.1265 - val_loss: 0.0147 - val_mae: 0.0923

Epoch 29/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0233 - mae: 0.1257

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0270 - mae: 0.1285 - val_loss: 0.0127 - val_mae: 0.0900

Epoch 30/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0172 - mae: 0.1132

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0311 - mae: 0.1373 - val_loss: 0.0124 - val_mae: 0.0880

Epoch 31/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0057 - mae: 0.0604

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0263 - mae: 0.1248 - val_loss: 0.0167 - val_mae: 0.1017

Epoch 32/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 27ms/step - loss: 0.0091 - mae: 0.0894

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0183 - mae: 0.1129 - val_loss: 0.0118 - val_mae: 0.0868

Epoch 33/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0132 - mae: 0.0933

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0231 - mae: 0.1201 - val_loss: 0.0135 - val_mae: 0.0922

Epoch 34/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 24ms/step - loss: 0.0145 - mae: 0.0986

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0152 - mae: 0.1014 - val_loss: 0.0213 - val_mae: 0.1140

Epoch 35/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0155 - mae: 0.1092

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0225 - mae: 0.1213 - val_loss: 0.0134 - val_mae: 0.0907

Epoch 36/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0157 - mae: 0.0969

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0208 - mae: 0.1079 - val_loss: 0.0127 - val_mae: 0.0878

Epoch 37/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0202 - mae: 0.1019

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0258 - mae: 0.1275 - val_loss: 0.0178 - val_mae: 0.1037

Epoch 38/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0110 - mae: 0.0912

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0197 - mae: 0.1088 - val_loss: 0.0180 - val_mae: 0.1042

Epoch 39/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0162 - mae: 0.1016

[1m10/10[0m [32m━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0229 - mae: 0.1259 - val_loss: 0.0145 - val_mae: 0.0943

Epoch 40/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0038 - mae: 0.0400

[1m10/10[0m [32m━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0169 - mae: 0.1022 - val_loss: 0.0256 - val_mae: 0.1241

Epoch 41/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0167 - mae: 0.1029

[1m10/10[0m [32m━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0212 - mae: 0.1121 - val_loss: 0.0127 - val_mae: 0.0882

Epoch 42/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 25ms/step - loss: 0.0229 - mae: 0.1237

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0161 - mae: 0.1037 - val_loss: 0.0176 - val_mae: 0.1038

Epoch 43/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 20ms/step - loss: 0.0204 - mae: 0.1149

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0183 - mae: 0.1109 - val_loss: 0.0139 - val_mae: 0.0925

Epoch 44/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0138 - mae: 0.0898

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0225 - mae: 0.1147 - val_loss: 0.0120 - val_mae: 0.0910

Epoch 45/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 22ms/step - loss: 0.0168 - mae: 0.0948

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0197 - mae: 0.1108 - val_loss: 0.0186 - val_mae: 0.1081

Epoch 46/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0213 - mae: 0.1263

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0248 - mae: 0.1202 - val_loss: 0.0127 - val_mae: 0.0894

Epoch 47/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 29ms/step - loss: 0.0316 - mae: 0.1306

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0211 - mae: 0.1139 - val_loss: 0.0132 - val_mae: 0.0905

Epoch 47: early stopping

Restoring model weights from the end of the best epoch: 32.

Training stopped at epoch 47

<Figure size 1200x400 with 1 Axes>

Figure saved as msft_training_loss.png

[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 107ms/step

```
[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 118ms/step

Test predictions shape : (23, 1)
Actual values shape    : (23, 1)

========================================
  SimpleRNN — Test Set Metrics
========================================
  MSE  : 549.3877
  RMSE : $23.44
  MAE  : $21.00
  MAPE : 5.21%
========================================

[1m1/3[0m [32m━━━━━━━━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 127ms/step

[1m3/3[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step

<Figure size 1400x600 with 1 Axes>

Figure saved as msft_rnn_predictions.png

<Figure size 600x600 with 1 Axes>

Figure saved as msft_scatter.png

[1mModel: "LSTM_Model"[0m
```

```
┏━━━
```

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m12s[0m 1s/step - loss: 0.7695 - mae: 0.8709

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m2s[0m 32ms/step - loss: 0.3743 - mae: 0.5808 - val_loss: 0.0251 - val_mae: 0.1325

Epoch 2/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0999 - mae: 0.3003

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0524 - mae: 0.1838 - val_loss: 0.1077 - val_mae: 0.2915

Epoch 3/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0313 - mae: 0.1539

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0216 - mae: 0.1185 - val_loss: 0.0342 - val_mae: 0.1577

Epoch 4/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 27ms/step - loss: 0.0201 - mae: 0.1258

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0234 - mae: 0.1200 - val_loss: 0.0528 - val_mae: 0.1935

Epoch 5/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 72ms/step - loss: 0.0177 - mae: 0.1035

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 11ms/step - loss: 0.0139 - mae: 0.0980 - val_loss: 0.0687 - val_mae: 0.2238

Epoch 6/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0040 - mae: 0.0531

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0109 - mae: 0.0861 - val_loss: 0.0534 - val_mae: 0.1948

Epoch 7/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0167 - mae: 0.0998

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0126 - mae: 0.0902 - val_loss: 0.0512 - val_mae: 0.1908

Epoch 8/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0108 - mae: 0.0783

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0159 - mae: 0.1023 - val_loss: 0.0532 - val_mae: 0.1947

Epoch 9/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 25ms/step - loss: 0.0069 - mae: 0.0523

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0115 - mae: 0.0816 - val_loss: 0.0540 - val_mae: 0.1964

Epoch 10/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0155 - mae: 0.0947

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0115 - mae: 0.0854 - val_loss: 0.0536 - val_mae: 0.1958

Epoch 11/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 23ms/step - loss: 0.0096 - mae: 0.0779

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0126 - mae: 0.0850 - val_loss: 0.0537 - val_mae: 0.1962

Epoch 12/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 26ms/step - loss: 0.0086 - mae: 0.0771

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 10ms/step - loss: 0.0129 - mae: 0.0884 - val_loss: 0.0446 - val_mae: 0.1793

Epoch 13/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 26ms/step - loss: 0.0099 - mae: 0.0714

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0100 - mae: 0.0803 - val_loss: 0.0512 - val_mae: 0.1919

Epoch 14/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 18ms/step - loss: 0.0061 - mae: 0.0622

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 7ms/step - loss: 0.0101 - mae: 0.0780 - val_loss: 0.0439 - val_mae: 0.1781

Epoch 15/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 19ms/step - loss: 0.0170 - mae: 0.1225

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 9ms/step - loss: 0.0126 - mae: 0.0881 - val_loss: 0.0492 - val_mae: 0.1884

Epoch 16/200

[1m 1/10[0m [32m━━━[0m[37m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m [1m0s[0m 21ms/step - loss: 0.0132 - mae: 0.0935

[1m10/10[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 8ms/step - loss: 0.0111 - mae: 0.0847 - val_loss: 0.0420 - val_mae: 0.1742

Epoch 16: early stopping

Restoring model weights from the end of the best epoch: 1.

LSTM training stopped at epoch 16

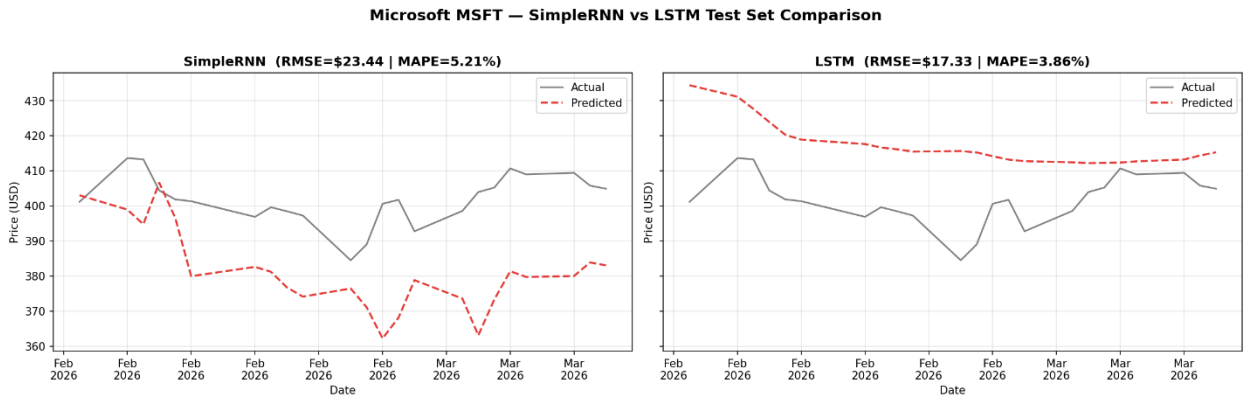
[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 125ms/step

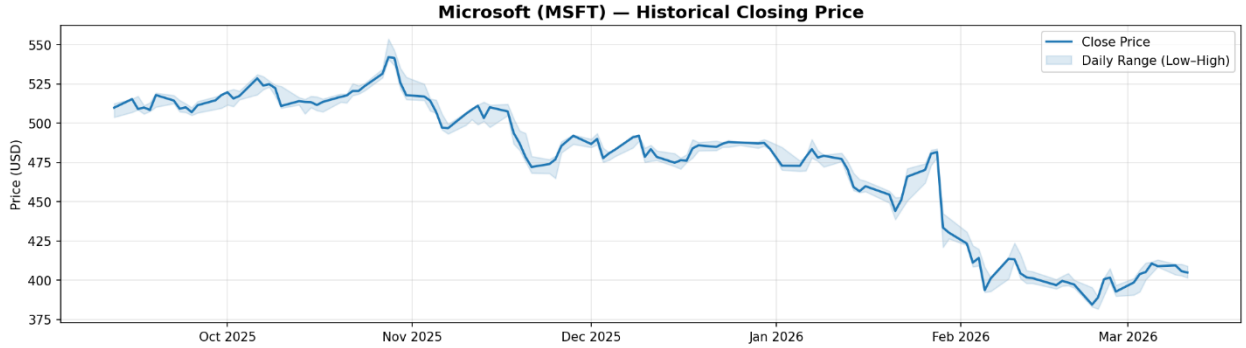
[1m1/1[0m [32m━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━[0m[37m[0m [1m0s[0m 140ms/step

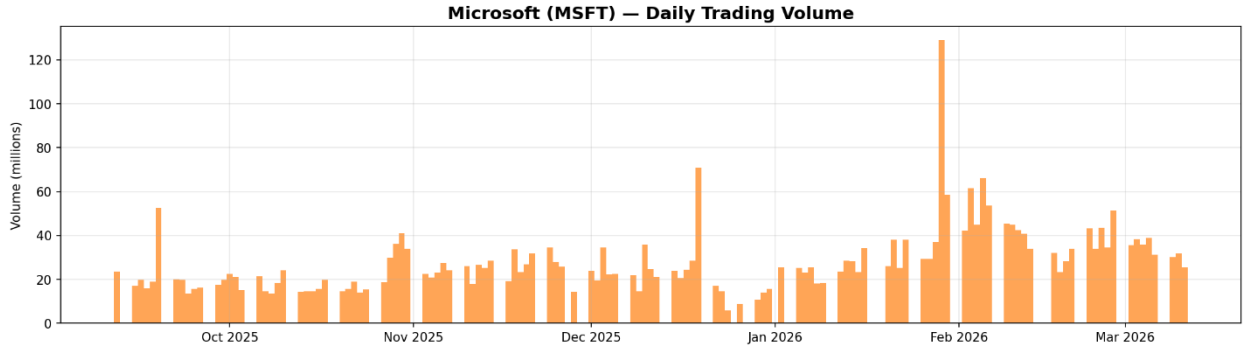
```
========================================
  LSTM — Test Set Metrics
========================================
  MSE  : 300.4501
  RMSE : $17.33
```

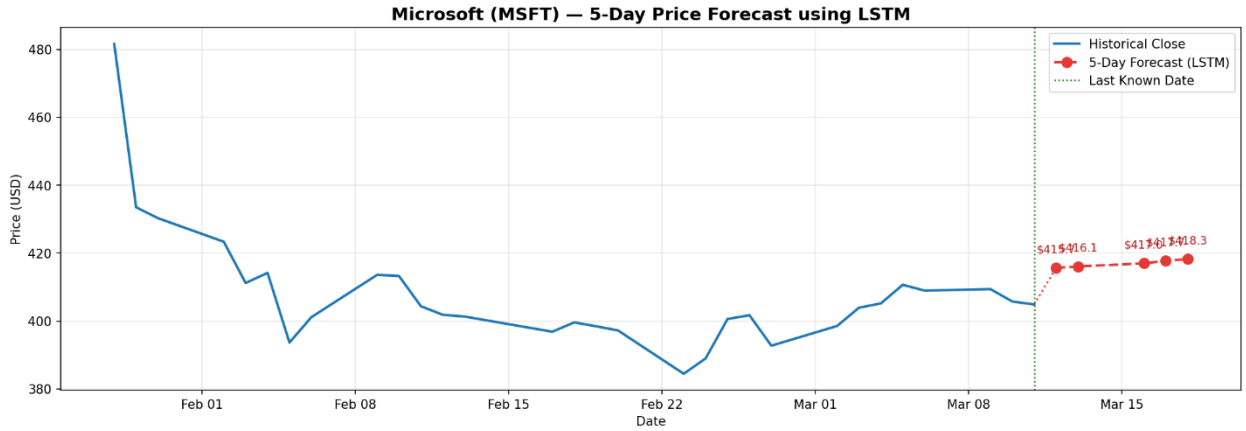
```
  MAE  : $15.40
  MAPE : 3.86%
========================================


<Figure size 1600x500 with 2 Axes>


Figure saved as msft_comparison.png


Best model selected: LSTM



────────────────────────────────────────────────────────
  5-Day MSFT Price Forecast (LSTM)
────────────────────────────────────────────────────────
  Thu 2026-03-12 : $415.65
  Fri 2026-03-13 : $416.08
  Mon 2026-03-16 : $416.99
  Tue 2026-03-17 : $417.71
  Wed 2026-03-18 : $418.27
────────────────────────────────────────────────────────


<Figure size 1400x500 with 1 Axes>


Figure saved as msft_forecast.png


    Model      MSE  RMSE ($)  MAE ($)  MAPE (%) Best?
SimpleRNN 549.3877     23.44     21.0      5.21
     LSTM 300.4501     17.33     15.4      3.86     ✓
```

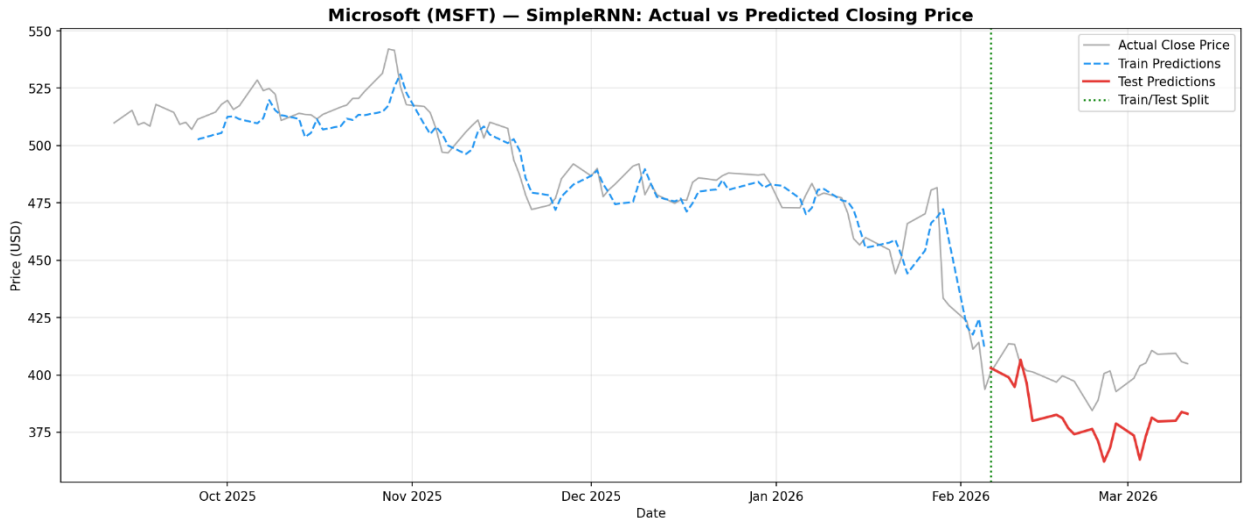
9. Visualization

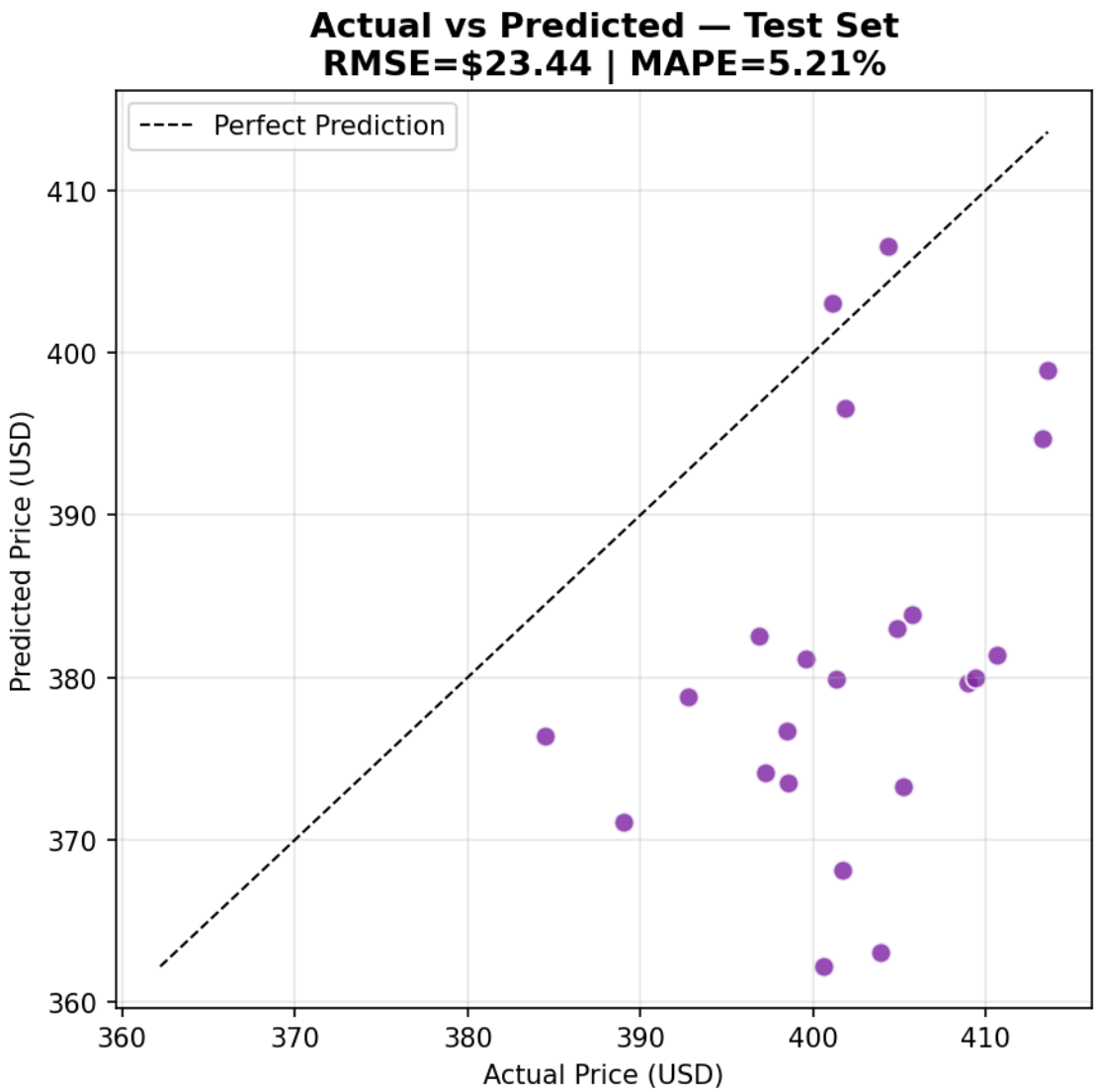


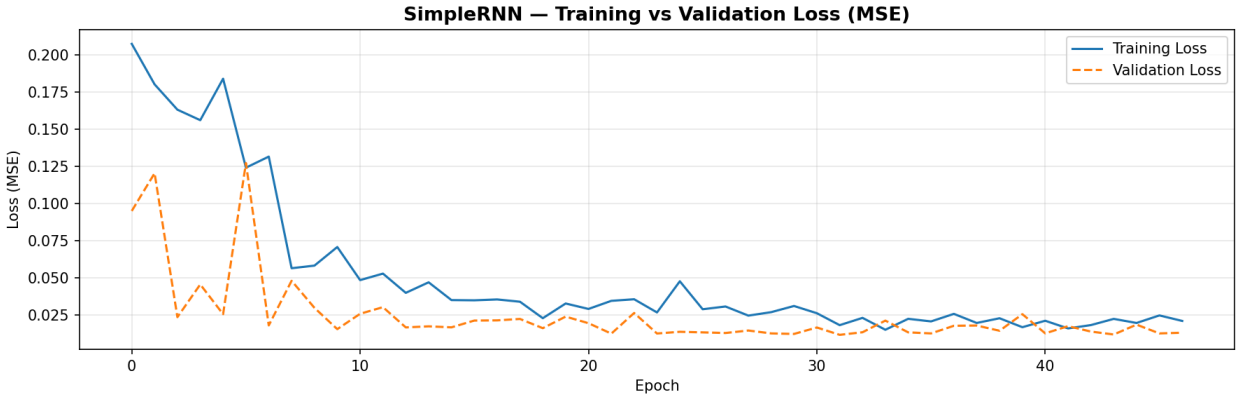







Microsoft (MSFT) — SimpleRNN: Actual vs Predicted Closing Price
Actual Close Price
Train Predictions
Test Predictions
Train/Test Split
Price (USD)
550
525
500
475
450
425
400
375
Oct 2025
Nov 2025
Dec 2025
Jan 2026
Feb 2026
Mar 2026
Date


Actual vs Predicted — Test Set
RMSE=$23.44 | MAPE=5.21%
Perfect Prediction
Predicted Price (USD)
410
400
390
380
370
360
360
370
380
390
400
410
Actual Price (USD)


SimpleRNN — Training vs Validation Loss (MSE)
Training Loss
Validation Loss
Loss (MSE)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0
10
20
30
40
Epoch